

Empirical Evaluation of the Follow-The-Regularized-Leader algorithm for the purpose of Online Portfolio Optimization

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ABSTRACT

- Online Portfolio Optimization involves strategically choosing the most efficient mix of diverse financial assets in real-time.
- This project focuses on implementing the Follow-The-Regularized-Leader (FTRL) algorithm to execute this task.
- This implementation of the algorithm involves applying this algorithm to a historical share price dataset
- The FTRL algorithm showcases its prowess by minimizing regrets, effectively reducing incurred losses.
- A comparative analysis is conducted between this algorithm and the FTL algorithm to evaluate their performance.

INTRODUCTION

- Portfolio optimization is the process of using mathematical approach to select the optimal combination of investment portfolio assets and their values, given a set of objectives and constraints in real-time.
- The goal of online portfolio optimization (OPO) is to reduce the loss (minimize risk), in turn maximizing the returns.
- The task is carried out via a variety of online optimization techniques, including FTL, FTRL, FTRL-p, Online Gradient Descent, Kalman Filters, etc.
- FTRL is an algorithm that balances the exploitation and exploration factor of the online algorithm resulting in optimized decisions for adversarial scenarios like the Portfolio Optimization.

DATASETS

- The Australian Historical Stock Prices dataset is used to implement this algorithm.
- The dataset contains historical share price data from top 100 companies listed on the Australian Securities Exchange.
- The dataset contains information including date, opening price, closing price, high price, low price, adjusted closing price and trading volume.
- For this task, we only use the date, opening price and closing price information.

METHODOLOGY

Follow-The-Regularized-Leader Algorithm:

- FTRL algorithm is a Follow-The-Leader (FTL) Algorithm with a regularization term added.
- FTL algorithm is an exploitative algorithm that purely uses the past data to make a decision.
- Adding a regularization term to the FTL algorithm helps achieve this task of decision making in adversarial applications.

Modelling the problem:

In the application of Online Portfolio Optimization, the problem is modelled as follows:

- Consider a scenario where we are playing the market for T days managing a portfolio consisting of i stocks.
- Every day we decide to buy or short certain number of shares of each stock. We make a decision to buy or short a number stocks every morning. At the end of the day, we sell whatever we bought and buy back the stock that we shorted.
- Let, w_t be a 1-D vector of size i , containing the decision. Here t indexes the day and every element represents the number of shares bought or sold of each stock.
- D represents the bound on the number of stocks that can be bought or shorted. In this implementation, $D=10$
- l_t represents the loss vector, that represents how much the stock depreciated. It is calculated as opening price – closing price.
- Now, the FTRL weight update is performed according to the equation:

$$\omega_t = \arg \min_{\omega \in B} \left[L_{t-1}(\omega) + \frac{1}{\eta} \|\omega\|_2^2 \right],$$

where, $L_{t-1}(\omega) = \sum_{s=1}^{t-1} l_s(\omega)$ and B is the l_∞ ball of D .

- And the Regret is calculated as:

$$R_T := H_T - L_T^*, \text{ where}$$

$$L_T^* := \min_{\omega \in B} \sum_{t=1}^T \langle \omega, l_t \rangle, \text{ and}$$

$$H_T := \sum_{t=1}^T \langle \omega_t, l_t \rangle$$

- L_T^* represents the loss obtained by using the best *fixed* decision in hindsight. H_T represents the total loss incurred by the FTRL algorithm.
- The value of η chosen for this application is dynamic, that is $\eta = D/G\sqrt{T}$ that achieves a low regret of order $O(DG\sqrt{T})$.

RESULTS

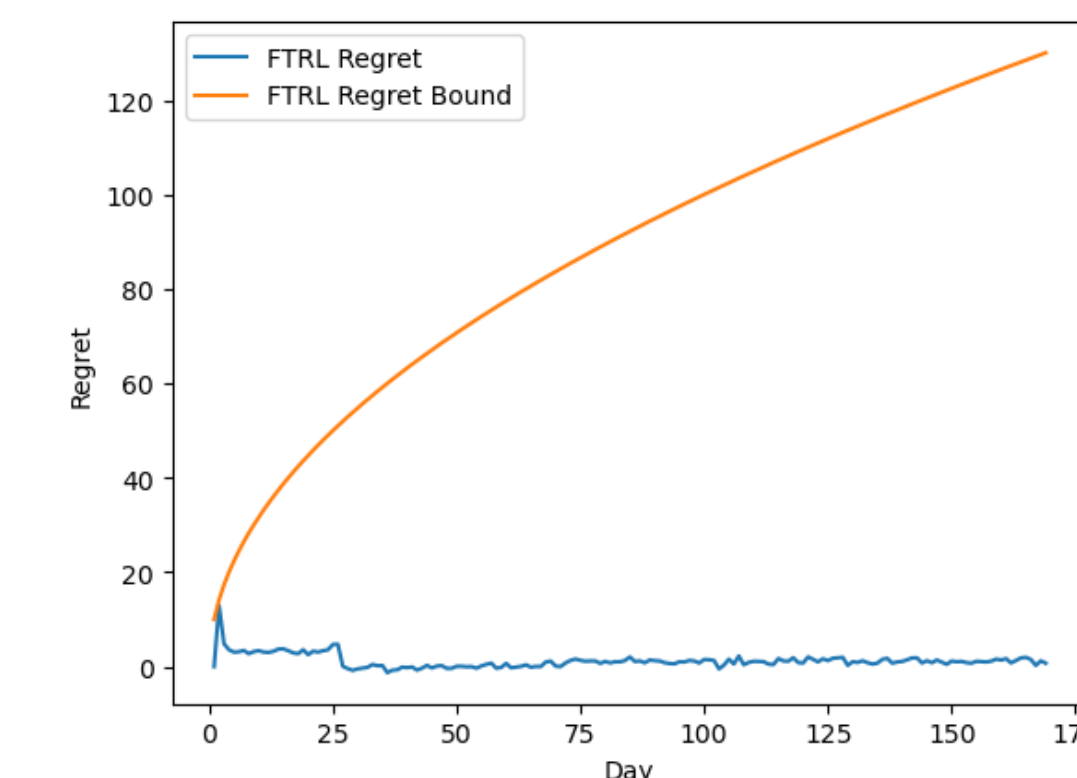


Fig 1: FTRL Regret graph

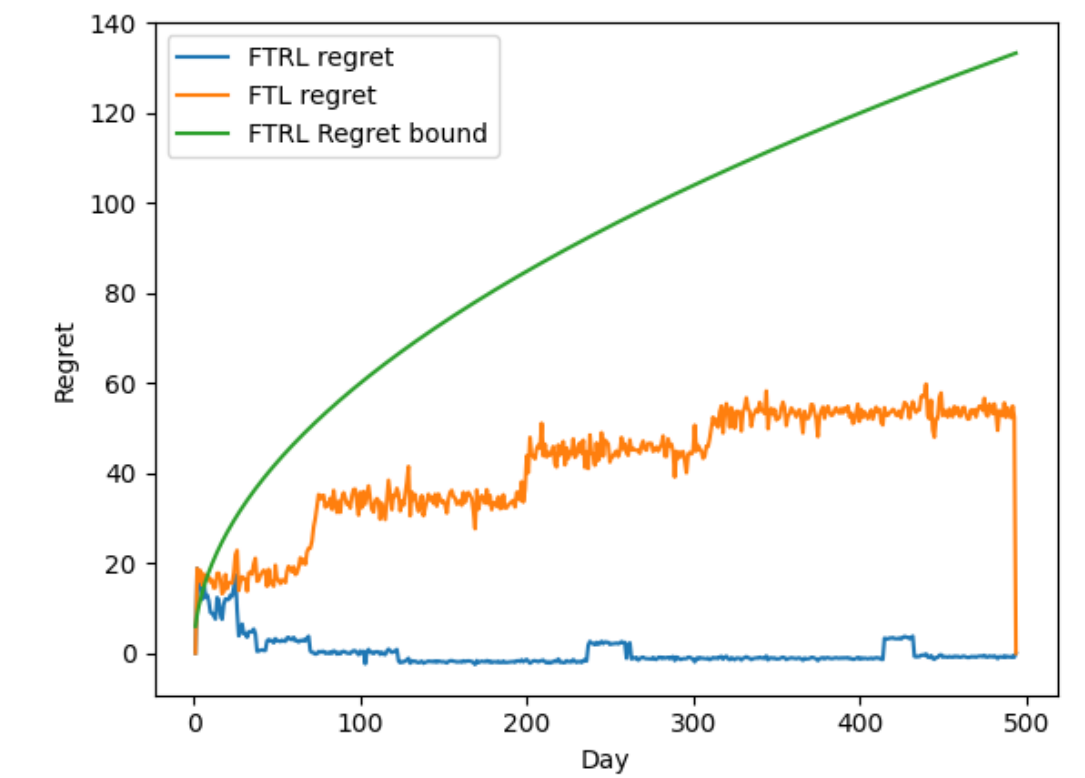


Fig 2: FTRL and FTL comparison

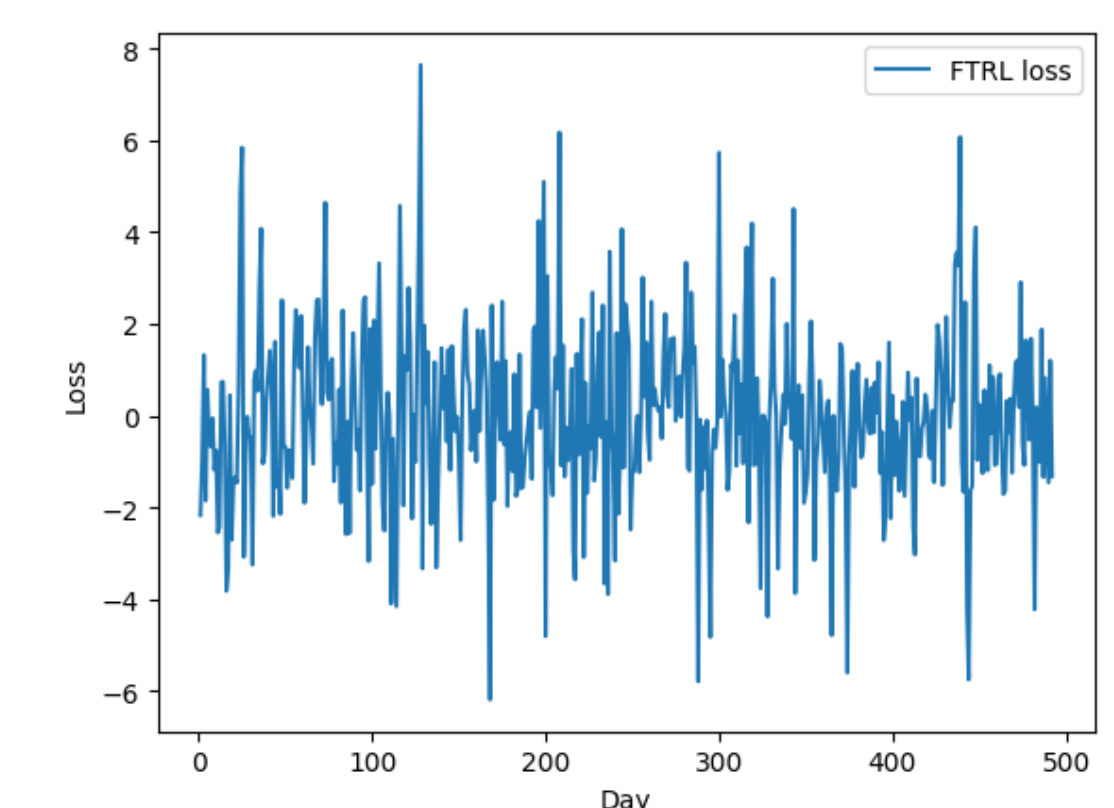


Fig 3: FTRL Loss graph

CONCLUSION

- Based on the results, it is evident that the FTRL algorithm surpasses the FTL algorithm in performance.
- The FTRL algorithm demonstrates notably lower regret compared to its FTL algorithm.
- The implementation adheres closely to the regret bound of the FTRL algorithm, noted as $O(DG\sqrt{T})$.
- The algorithm effectively minimizes loss, as observed by the limited increase in loss, which does not exceed 4 points.
- However, FTRL algorithm becomes computationally expensive as the number of financial instruments increases.

REFERENCES

- <https://courses.cs.washington.edu/courses/cse599s/14sp/scribes/lecture3/lecture3.pdf>
- Amit Agarwal, Elad Hazan, Satyen Kale, and Robert E. Schapire. 2006. Algorithms for portfolio management based on the Newton method. In Proceedings of the 23rd international conference on Machine learning (ICML '06). Association for Computing Machinery, New York, NY, USA, 9–16. <https://doi.org/10.1145/1143844.1143846>
- Lecture notes (Lecture 3,4,8 and 9)